



HCIG Intro and Basics

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Announcements



- Networking and Linux Security
 - 11am – 12pm Bruner 327 Sept. 3rd
- Joint networking meeting
 - 6pm – 7pm Bruner 228 Sept 3rd



What is Hardware Security?



Software security usually assumes the chip underneath is telling the truth. Hardware security is what happens when that assumption breaks. Typically when an attacker gets physical or electrical access to a device's chips, buses, and pins, not just its code.

SOFTWARE ATTACK SURFACE

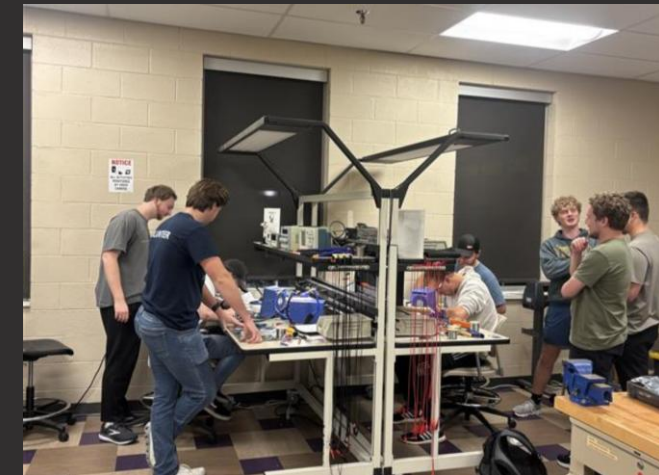
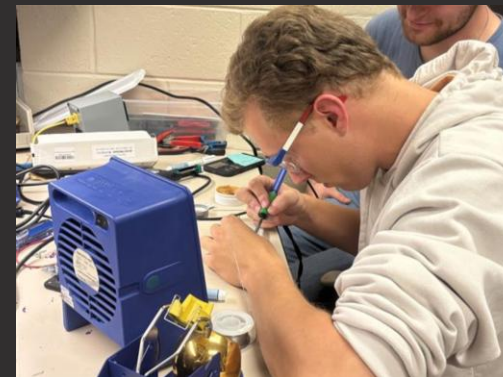
- Application code & logic
- Operating system
- Network protocols
- APIs & user input

HARDWARE ATTACK SURFACE

- Debug ports (JTAG / UART)
- Power lines & clock signals
- On-board buses (I2C / SPI)
- Chip packaging & EM emissions

MITRE eCTF

- Teams design a secure embedded system to spec, then attack every other team's build
- Runs on an actual physical board
- Design Phase rewards secure engineering; Attack Phase rewards breaking others' designs



Lab #1



- Firmware Flashing and UART Communication
- Learn how to flash an ESP32-C3 microcontroller



Hardware Security Examples



- Teraleak Pokémon Legends Z-A
 - Teraleak was hacked and secretly hackers gained access to the Generation 10 game Legends Z-A. An early leaked version of the game was released.
- ATM Jackpotting Demo
 - Jack Barnaby at Black Hat 2010 demoed a custom code upload to a normal ATM that would dispense bills whenever he would have liked.
- Xbox 360 JTAG Hack
 - Soldering onto exposed debug pins would bypass the console's code-signing process and run unsigned software

Hardware Security Jobs



Embedded / Firmware Security Engineer

Builds the secure boot, encryption, and update mechanisms that run directly on a device's chip.



Hardware Reverse Engineer

Tears down chips and boards to find out how they really work and where they're weak.



Side-Channel & Fault Injection Analyst

Tests whether a chip leaks secrets through power or timing, or can be glitched into misbehaving.



IoT / Hardware Penetration Tester

Gets hands-on with real devices probing debug ports, sniffing buses, hunting exploitable flaws.



Cryptographic Hardware Engineer

Designs the crypto circuits and hardware roots of trust that protect data at the silicon level.



Defense & Aerospace Systems Security Engineer

Protects mission hardware from weapons systems to secure comms against tampering.

Lab #2



- Walkthrough of Microcorruption embedded assembly
- <https://microcorruption.com/>
- Feel free to follow along!



Why This Matters



- Every layer above the transistor assumes the transistor is honest
 - Operating systems, apps, and networks all inherit whatever trust (or mistrust) the hardware underneath actually deserves.
- This is where embedded CTFs live
 - Competitions like MITRE eCTF put exactly these ideas into practice teams design secure embedded systems, then attack each other's.
- It's the meeting point of two disciplines
 - Hardware security sits right at the intersection of computer engineering and cybersecurity which makes it a great home base for this club.

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